

SAMUTPRAKAN AGROMET, Thailand

WMO: 484200

Lat: 13.517N

Lon: 100.762E

Elev: 3

StdP: 101.29

Time Zone: 7.00 (E07)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.6	21.2	12.8	9.2	22.2	14.7	10.4	24.1	8.2	25.2	7.3	26.8	1.8	40	0.409

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	4.2	33.7	26.4	33.2	26.5	32.8	26.5	28.4	32.0	28.1	31.7	27.7	31.3	3.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.3	23.2	31.3	27.0	22.7	30.9	26.7	22.3	30.5	91.7	32.1	90.0	31.8	88.7	31.5	29.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.8	6.9	6.2	17.9	35.4	1.8	1.3	16.6	36.3	15.6	37.0	14.5	37.8	13.2	38.7		
			14.3	29.0	2.1	0.6	12.8	29.5	11.5	29.8	10.3	30.1	8.8	30.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.8	26.8	28.1	29.2	30.1	30.2	29.6	29.1	29.1	28.8	28.8	28.4	27.1
	DBStd	1.65	1.90	1.25	1.33	1.20	1.21	0.85	0.94	0.88	0.95	1.09	1.35	2.06
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6854	521	508	594	604	627	587	593	591	564	583	553	529
	CDD18.3	3812	262	274	336	354	368	337	334	333	314	325	303	271
	CDH23.3	47069	2603	3121	4212	4787	5086	4577	4321	4292	3888	3895	3498	2789
Wind	WSAvg	2.7	2.0	2.9	3.5	3.5	3.1	3.5	3.3	3.1	2.4	1.7	1.9	2.0
	PrecAvg	1485	17	17	49	84	194	178	168	197	304	230	39	8
Precipitation	PrecMax	1944	87	65	154	219	396	324	267	347	470	419	142	42
	PrecMin	1016	0	0	5	14	61	60	44	94	203	56	0	0
	PrecStd	222	21	17	42	61	83	66	60	77	64	78	38	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.7	33.2	33.7	34.1	34.2	33.2	32.6	33.1	33.6	33.6	33.7	33.9
		MCWB	24.2	23.4	24.8	26.7	27.4	27.2	26.3	26.7	26.6	26.4	25.7	25.4
	2%	DB	31.7	32.0	32.4	33.3	33.4	32.5	32.0	31.9	32.4	32.9	32.9	32.8
		MCWB	23.8	23.8	25.9	27.1	27.4	27.0	26.5	26.5	26.8	26.3	25.5	25.0
	5%	DB	31.0	31.1	31.6	32.7	33.0	32.1	31.5	31.5	31.7	32.2	32.3	31.8
		MCWB	24.2	25.1	26.4	27.2	27.4	26.9	26.4	26.4	26.6	26.3	25.6	24.8
	10%	DB	30.3	30.5	31.2	32.2	32.4	31.6	31.1	31.0	31.2	31.5	31.6	30.9
		MCWB	24.3	25.3	26.5	27.3	27.2	26.8	26.3	26.3	26.4	26.2	25.5	24.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.0	27.2	28.0	28.9	29.0	28.2	27.6	27.7	28.0	27.7	27.7	27.2
		MCDB	30.1	30.4	31.5	32.5	32.7	31.7	30.5	30.9	31.9	31.6	31.3	30.8
	2%	WB	26.5	26.7	27.6	28.4	28.5	27.7	27.2	27.2	27.4	27.2	27.2	26.6
		MCDB	29.5	29.9	31.0	32.1	32.1	31.2	30.5	30.6	31.0	30.9	30.8	30.0
	5%	WB	26.1	26.5	27.2	28.1	28.1	27.4	27.0	27.0	27.1	27.0	26.8	26.2
		MCDB	29.1	29.7	30.6	31.7	31.7	30.9	30.4	30.4	30.5	30.6	30.4	29.6
	10%	WB	25.7	26.2	27.0	27.7	27.7	27.2	26.7	26.7	26.7	26.7	26.5	25.7
		MCDB	28.6	29.3	30.4	31.3	31.4	30.7	30.2	30.2	30.1	30.1	29.9	29.2
Mean Daily Temperature Range	5% DB	MDBR	6.8	5.0	4.1	4.1	4.2	3.9	3.7	3.9	4.6	5.2	5.9	7.0
		MCDBR	7.4	6.2	4.8	4.4	4.5	4.6	4.0	4.3	5.3	5.9	6.6	7.6
	5% WB	MCWBR	2.6	2.4	2.2	2.1	2.0	2.0	1.9	2.0	2.2	2.0	2.1	2.5
		MCDBR	4.9	3.9	3.7	3.6	4.0	4.4	3.9	4.2	5.0	5.5	5.7	6.0
		MCWBR	2.0	1.8	2.0	2.0	2.1	2.1	2.0	2.1	2.2	2.2	2.1	2.3
Clear-Sky Solar Irradiance	taub		0.542	0.608	0.630	0.570	0.496	0.467	0.461	0.463	0.484	0.570	0.518	0.503
	taud		1.970	1.835	1.819	2.003	2.264	2.345	2.361	2.343	2.272	1.987	2.114	2.108
	Ebn at Noon		759	727	724	766	811	826	833	841	829	748	776	783
	Edh at Noon		179	212	220	183	138	126	124	128	138	180	154	153
All-Sky Solar Radiation	RadAvg		5.10	5.50	5.61	5.98	5.42	5.18	4.96	5.03	4.87	4.76	4.92	4.92
	RadStd		0.30	0.24	0.40	0.38	0.37	0.26	0.26	0.31	0.27	0.26	0.34	0.24

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (31 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.20	+0.21	N/A	N/A	+70

Nomenclature: See separate page